LOW AND REVERSE BRAKE (PISTON A) CLEARANCE MEASUREMENT/ADJUSTMENT

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Preparation Before Servicing

1. Print out the measurement/adjustment value input sheet. (See DIFFERENTIAL BACKLASH MEASUREMENT/ADJUSTMENT.)

Note

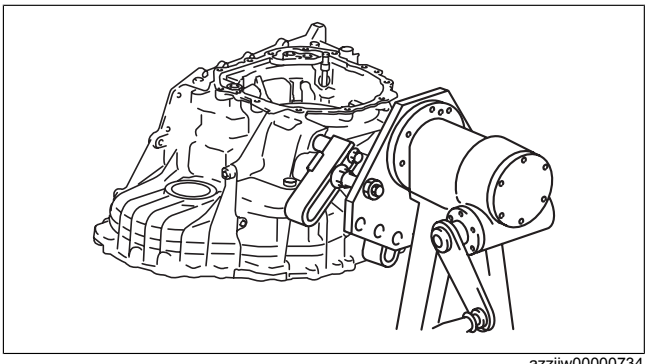
- When performing the measurement/adjustment, input the measured and calculated values into the measurement/adjustment value input sheet.
- If the measurement/adjustment value input sheet has already been printed out for the other measurements/adjustments, use the sheet.

Low and Reverse Brake (Piston A) Clearance Measurement

1. Rotate and adjust the rotation handle of the engine stand so that the end cover side is facing upward.

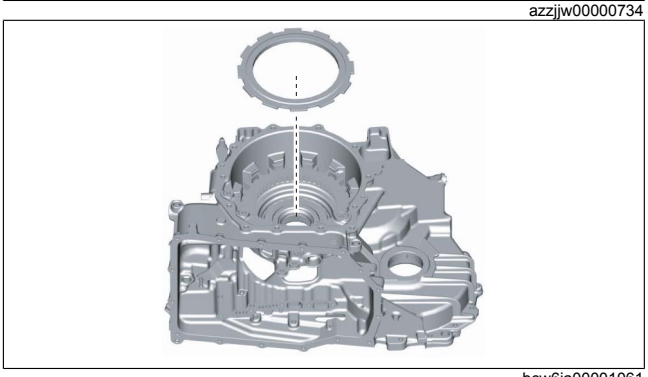
Caution

- **To reduce measurement differences during the low and reverse brake (piston A) clearance measurement, adjust so that the alignment surface of the transaxle case with the end cover is level.**

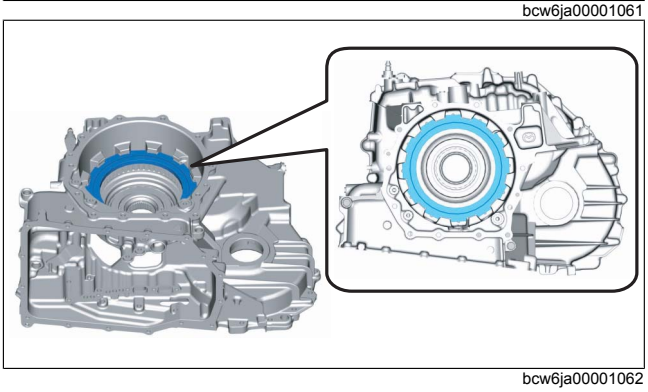


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2. Assemble the retaining plate.



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3. Assemble the snap ring using the following procedure:

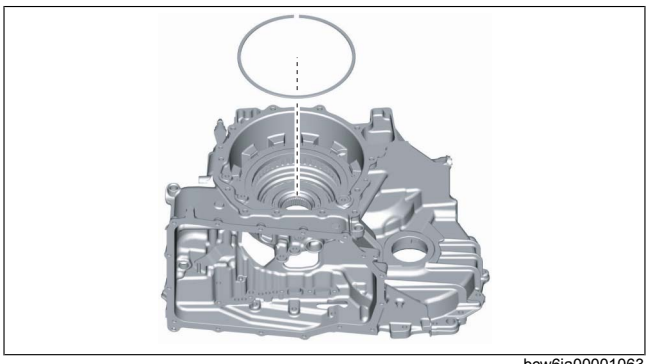
Note

- Snap ring size: Outer diameter approx. 191.1 mm {7.524 in}

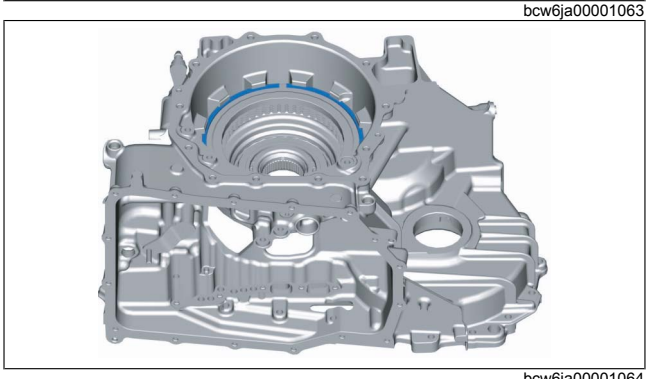
(1) Assemble the snap ring to the position shown in the figure.

Caution

- **After assembling the snap ring, verify that the snap ring is securely inserted into the bottom of the snap ring groove.**

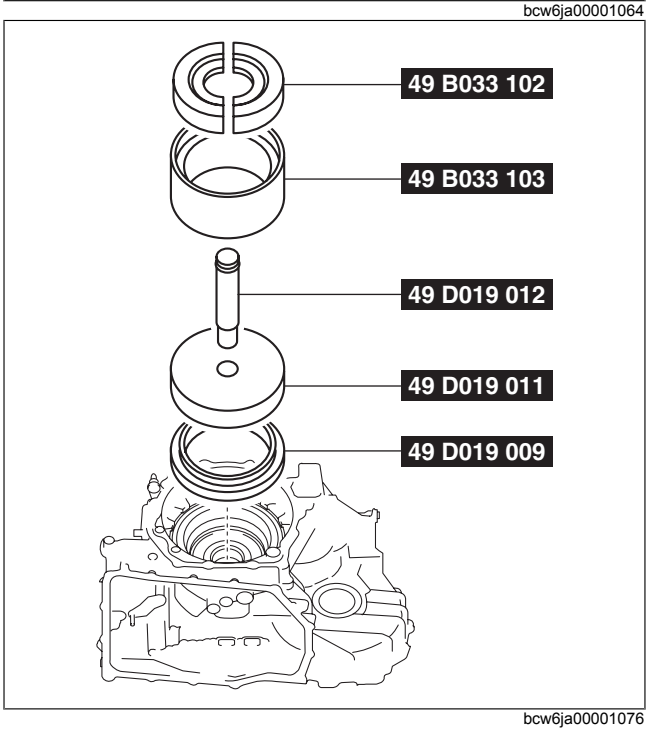


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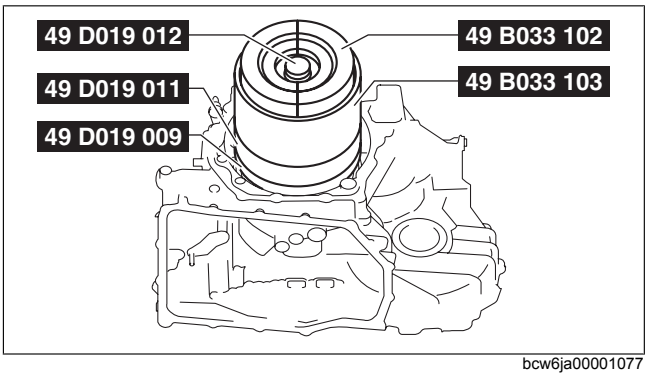


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4. Install the SSTs.



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5. Set the measuring instrument to the transaxle case using the following procedure.
   (1) Install the SST for securing the magnetic stand using the procedure shown in the figure.

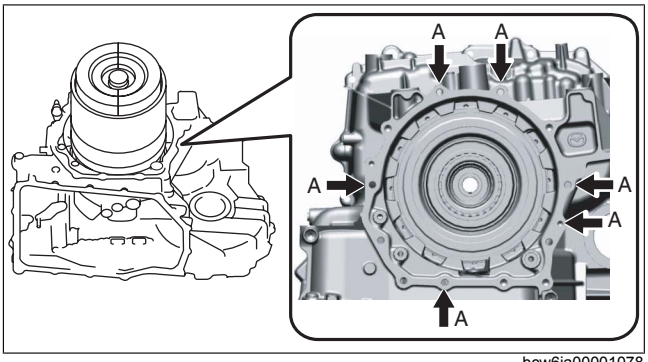
Caution
- **If the bolt is tightened with excessive force when installing the SST, the alignment surface of the end cover with the transaxle case could be damaged. Tighten the bolt so that the SST does not move during low and reverse brake (piston A) clearance measurement.**
- **To prevent damage to the parts, the bolt holes in the transaxle case which are used for securing the SST are the penetrated bolt holes shown in the figure.**

Note
- When installing the SST to the transaxle case, use bolts (M8 × 1.25).
- Use a bolt length in which the bolt end comes out of the transaxle case bolt hole but does not contact the transaxle case.

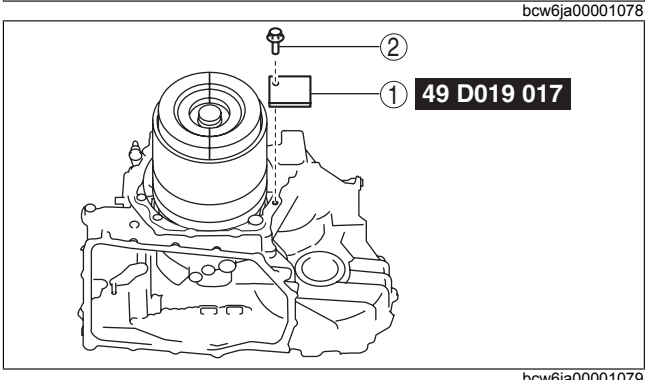
A : Bolt installation position for SST installation

* : Use a bolt length in which the bolt end comes out of the transaxle case bolt hole but does not contact the transaxle case.

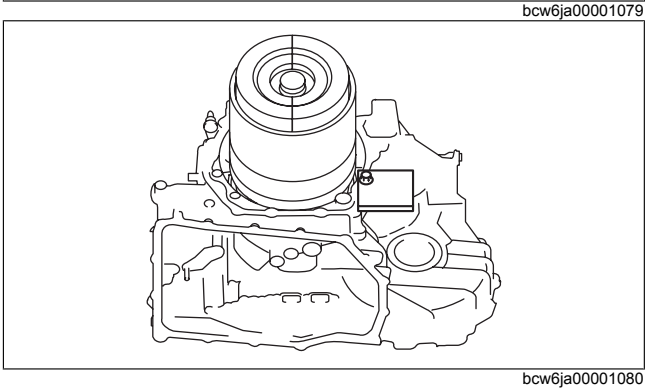
SST installation bolt tightening torque
15 N·m {1.5 kgf·m, 11 ft·lbf} or less (tighten so that SST does not move during low and reverse brake (piston A) clearance measurement)



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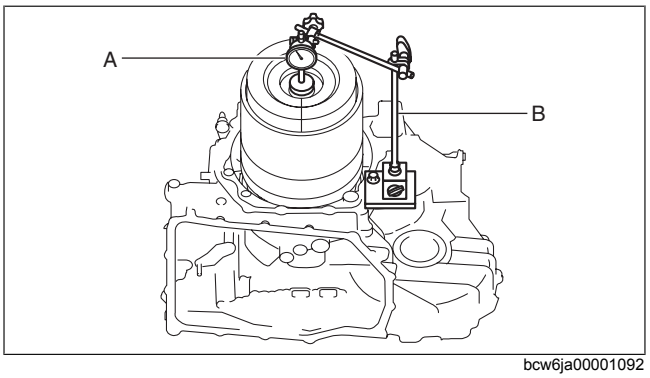

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1	SST (for securing magnetic stand)
2	Bolt (M8 × 1.25)*

(2) Set the dial gauge and magnetic stand as shown in the figure.

Caution
- **To reduce measurement differences during the low and reverse brake (piston A) clearance measurement, set the dial gauge so that it is perpendicular to the alignment surface of the transaxle case with the end cover.**

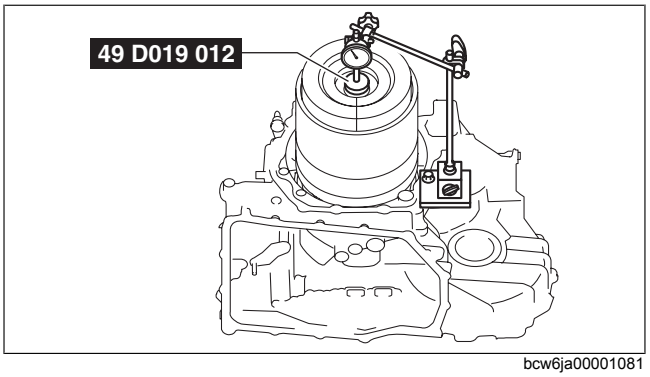
A : Dial gauge
B : Magnetic stand



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(3) Set the dial gauge end to the center of the **SST** (49 D019 012).

Note
- Measure the stroke amount of low and reverse brake piston A as the low and reverse brake (piston A) clearance.



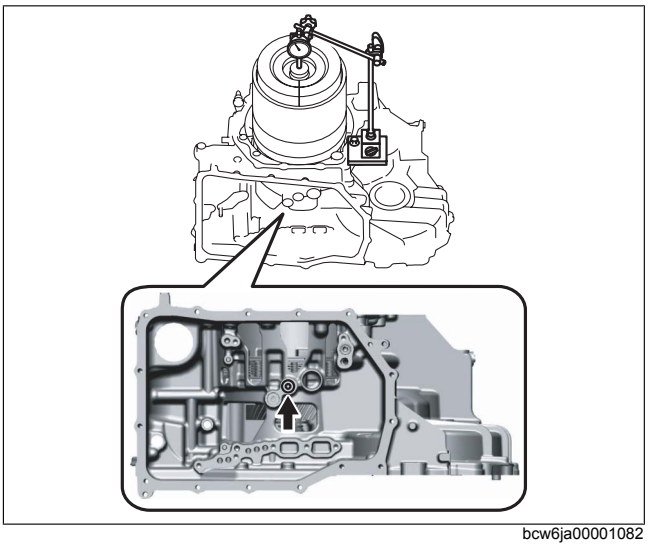
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6. Measure the low and reverse brake (piston A) clearance using the following procedure:
   (1) Blow compressed air into the oil passage shown in the figure to force low and reverse brake piston A to stroke (approx. 3 times).

Warning
- **Always wear protective eye wear when using the air compressor. Otherwise, ATF or dirt particles blown off by the air compressor could get into the eyes.**

Caution
- **To prevent damage to parts, always use an air compressor which is adjusted to the indicated pressure.**

Compressed air pressure
0.39—0.44 MPa {4.0—4.4 kgf/cm2, 57—63 psi}

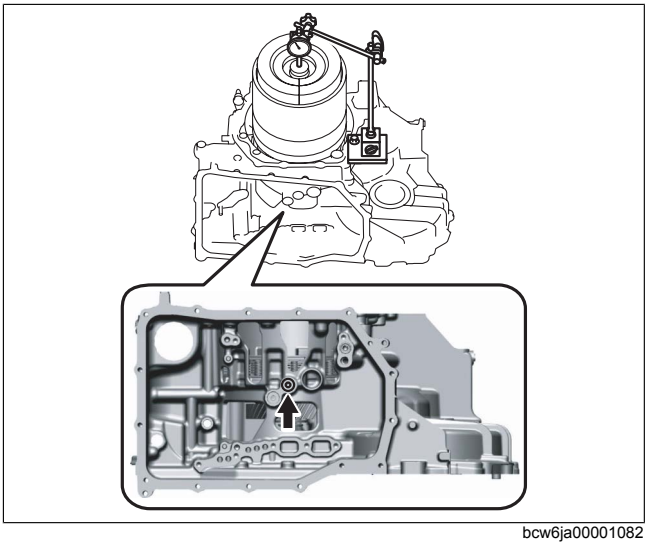

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(2) Blow compressed air into the oil passage shown in the figure to operate low and reverse brake piston A, and read the value when the dial gauge is stabilized.

Warning
- **Always wear protective eye wear when using the air compressor. Otherwise, ATF or dirt particles blown off by the air compressor could get into the eyes.**

Caution
- **To prevent damage to parts, always use an air compressor which is adjusted to the indicated pressure.**
- **To reduce measurement differences during the low and reverse brake (piston A) clearance measurement, gently blow and release compressed air.**

Compressed air pressure
0.39—0.44 MPa {4.0—4.4 kgf/cm2, 57—63 psi}


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(3) Input the dial gauge value, which was read while low and reverse brake piston A was operating, into the measurement/adjustment value input sheet.
(4) Release the compressed air and read the dial gauge value while low and reverse brake piston A is not operating.
(5) Input the dial gauge value, which was read while low and reverse brake piston A was not operating, into the measurement/adjustment value input sheet.
(6) Perform the following calculation to calculate the low and reverse brake (piston A) clearance.
Low and reverse brake (piston A) clearance = A - B
A: Dial gauge value while low and reverse brake piston A is operating
B: Dial gauge value while low and reverse brake piston A is not operating

Note
Example
A: Dial gauge value while low and reverse brake piston A is operating is 3.060 mm {0.1205 in}
B: Dial gauge value while low and reverse brake piston A is not operating is 0.595 mm {0.0234 in}
Low and reverse brake (piston A) clearance = 3.060 mm {0.1205 in} - 0.595 mm {0.0234 in} = 2.465 mm {0.09705 in}

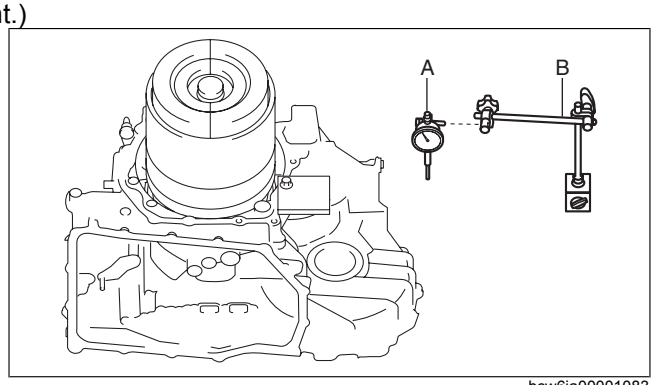
(7) Input the calculated low and reverse (piston A) clearance into the measurement/adjustment value input sheet.
(8) Verify that the low and reverse brake (piston A) clearance satisfies the specification.

Specification
2.130—2.330 mm {0.08386—0.09173 in}

- If not within the specification, adjust the low and reverse brake (piston A) clearance. (See Low and Reverse Brake (Piston A) Clearance Adjustment.)

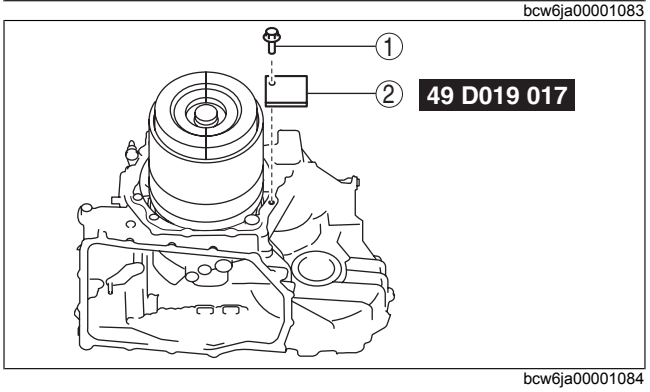
7. Remove the dial gauge and magnetic stand.

A : Dial gauge
B : Magnetic stand



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8. Remove the SST for securing the magnetic stand using the procedure shown in the figure.



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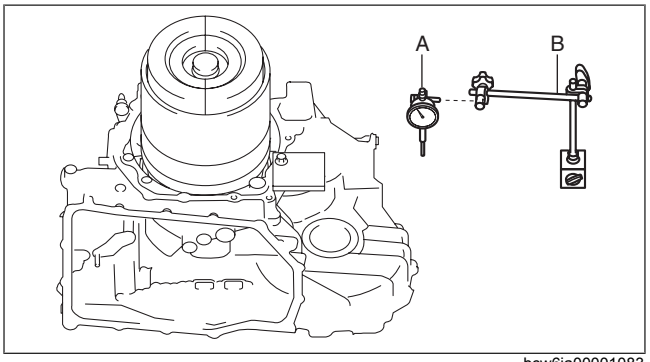
1	Bolt
2	SST (for securing magnetic stand)

9. Remove the SSTs.

Low and Reverse Brake (Piston A) Clearance Adjustment

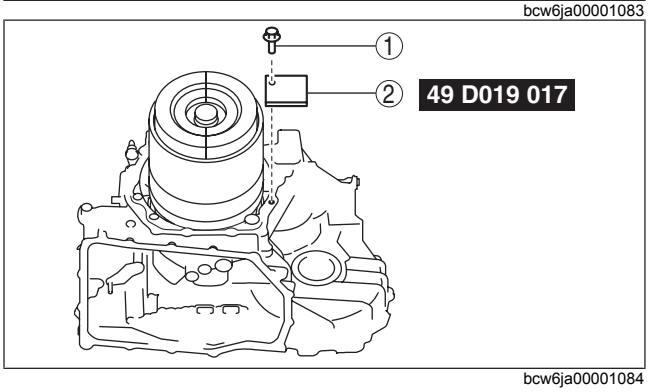
1. Remove the dial gauge and magnetic stand.

A : Dial gauge
B : Magnetic stand



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2. Remove the SST for securing the magnetic stand using the procedure shown in the figure.

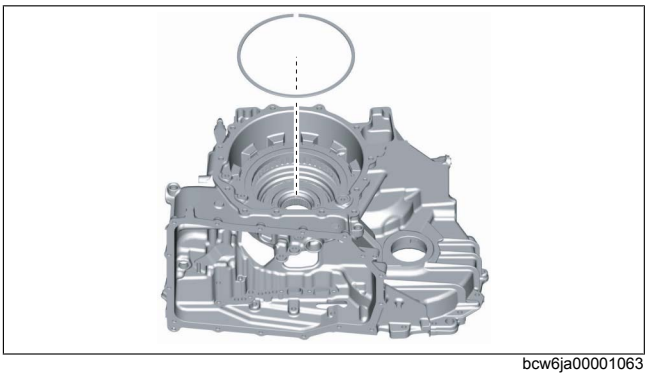


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1	Bolt
2	SST (for securing magnetic stand)

3. Remove the SSTs.

4. Remove the snap ring.
5. Measure the thickness of the removed snap ring.


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Note
- Recommended measuring instrument: Micrometer

6. Input the measured snap ring thickness into the measurement/adjustment value input sheet.
7. Perform the following calculation to calculate the low and reverse brake (piston A) clearance gap.

Note
- The low and reverse brake (piston A) clearance gap is the difference between the low and reverse brake (piston A) clearance and the median value of the low and reverse brake (piston A) clearance specification.

Low and reverse brake (piston A) clearance gap = C-G
C: Low and reverse brake (piston A) clearance
G: median value of low and reverse brake (piston A) clearance specification (2.230 mm {0.08780 in})

Note
Example
C: Low and reverse brake (piston A) clearance is 2.465 mm {0.09705 in}
Low and reverse brake (piston A) clearance gap = 2.465 mm {0.09705 in} - 2.230 mm {0.08780 in} = 0.235 mm {0.00925 in}

8. Input the calculated low and reverse brake (piston A) clearance gap into the measurement/adjustment value input sheet.
9. Perform the following calculation to calculate the optimum snap ring thickness.
Optimum snap ring thickness = F + H
F: Thickness of removed snap ring
H: Low and reverse brake (piston A) clearance gap

Note
Example
F: Thickness of removed snap ring is 2.125 mm {0.08366 in}
Thickness of optimum snap ring = 2.125 mm {0.08366 in} + 0.235 mm {0.00925 in} = 2.360 mm {0.09291 in}

10. Input the calculated optimum snap ring thickness into the measurement/adjustment value input sheet.
11. From the following table, select a snap ring of a thickness closest to the calculated optimum snap ring thickness.

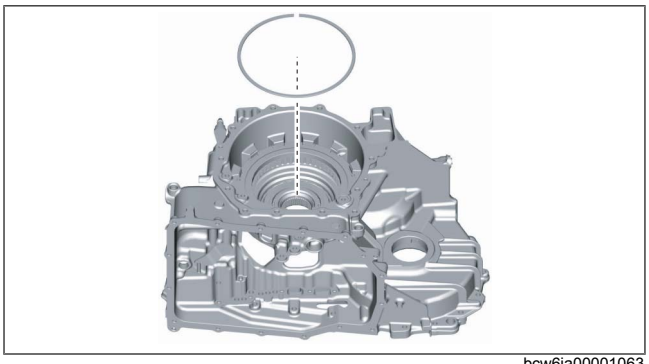
Snap ring thickness
2.7 mm {0.11 in}
2.6 mm {0.10 in}
2.5 mm {0.098 in}
2.4 mm {0.094 in}
2.3 mm {0.091 in}
2.2 mm {0.087 in}
2.1 mm {0.083 in}
2.0 mm {0.079 in}
1.9 mm {0.075 in}
1.8 mm {0.071 in}

12. Assemble the selected snap ring using the following procedure:

Note
- Snap ring size: Outer diameter approx. 191.1 mm {7.524 in}

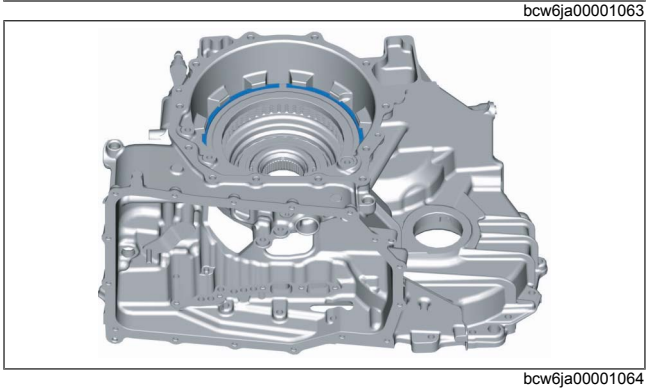
(1) Assemble the snap ring to the position shown in the figure.

Caution
- **After assembling the snap ring, verify that the snap ring is securely inserted into the bottom of the snap ring groove.**



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13. Perform the low and reverse brake (piston A) clearance measurement from Step 4. (See Low and Reverse Brake (Piston A) Clearance Measurement.)



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